UNIVERSITÀ DEGLI STUDI DI MILANO

FACOLTÀ DI SCIENZE POLITICHE,
ECONOMICHE E SOCIALI

Convolutional Neural Network Architectures for Rock–Paper–Scissors Image Classification

End-Of-Course Project

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Data Science for Economics (Master's Degree)
II Year

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February 28, 2026

Introduction

Convolutional Neural Networks (CNNs) have become a standard approach for image classification tasks due to their ability to automatically learn hierarchical feature representations from raw visual data. In recent years, CNN-based models have demonstrated strong performance across a wide range of computer vision applications, from object recognition to gesture and action classification.

This project investigates the application of convolutional neural networks to the classification of hand gesture images representing the Rock–Paper–Scissors game. Using a curated image dataset, three CNN architectures of increasing complexity are designed, trained, and systematically compared under identical experimental conditions. The analysis focuses not only on classification accuracy, but also on training dynamics, error patterns, and robustness to domain shift, providing a comprehensive evaluation of how architectural choices influence predictive performance and generalization.

1 Data Exploration and Preprocessing

1.1 Dataset Description and Exploratory Analysis

The dataset used in this project was obtained from a publicly available Kaggle repository (`https://www.kaggle.com/datasets/drgfreeman/rockpaperscissors`) containing images of hand gestures for the Rock–Paper–Scissors game. The dataset consists of a total of 2,188 RGB images, evenly distributed across the three target classes: rock (726 images), paper (710 images), and scissors (752 images). All images were captured against a green background under relatively consistent lighting and white balance conditions, resulting in a visually homogeneous and well-curated dataset.

Each image is stored in PNG format with an original resolution of 300 × 200 pixels and is organized into class-specific subdirectories, enabling a straightforward mapping between file structure and class labels. An initial exploratory analysis confirmed the near-balanced class distribution and the absence of missing or corrupted files. Random visual inspection of samples from each class further revealed that the hand gestures are generally well-centered and clearly distinguishable, with limited intra-class variability compared to real-world acquisition scenarios. This level of consistency makes the dataset particularly suitable for controlled experiments focused on architectural comparisons rather than robustness to extreme visual noise.

To avoid any form of data leakage, external datasets are introduced only at a later stage of the analysis. These datasets are used exclusively for evaluation and play no role in model training or model selection, allowing generalization to be assessed independently of the training distribution.

1.2 Train – Validation Split

The dataset was split into training and validation subsets using a stratified sampling strategy, with 80% of the images allocated to training and 20% reserved for validation. Stratification was performed at the file level to preserve the original class proportions in both splits and to guarantee that all classes are represented in each subset.

A fixed random seed was used to ensure reproducibility of the split across runs. The validation set was held out entirely from training and used exclusively for model selection, hyperparameter tuning, and performance evaluation. This splitting strategy provides a reliable estimate of in-distribution generalization while maintaining methodological transparency and repeatability.

1.3 Image Preprocessing: Resizing and Normalization

After splitting the dataset into training and validation subsets, a common preprocessing pipeline was applied to all images: to ensure compatibility with convolutional neural network architectures and to standardize the input space, images were resized to a fixed square resolution of 96 × 96 pixels. Resizing was performed using bilinear interpolation with antialiasing, preserving the essential spatial structure of the hand gestures while reducing computational cost and memory requirements.

Following resizing, pixel intensities were converted to floating-point format and normalized to the [0, 1] range, with the aim of facilitating stable and efficient optimization by preventing large gradient magnitudes during training. This normalization step was applied directly within the data pipeline rather than as a dedicated rescaling layer inside the models, ensuring consistent preprocessing across all architectures and simplifying model definitions.

1.4 Data Augmentation

Although the dataset is visually consistent and relatively clean, light data augmentation was applied during training to encourage better generalization and to reduce sensitivity to small spatial variations. Augmentation operations were deliberately kept mild to avoid introducing unrealistic distortions, given the already constrained nature of the dataset.

Specifically, the training pipeline includes random horizontal flips, small rotations, zooming, translations, and slight contrast adjustments. These transformations are applied on-the-fly and exclusively to the training set, ensuring that validation data remains untouched and representative of the original distribution. This strategy allows the models to learn more robust feature representations while preserving a fair and unbiased evaluation protocol.

2 CNN Architecture and Training

2.1 Design of CNN Architectures with Incremental Complexity

The three convolutional neural network architectures evaluated in this study were explicitly designed to form a controlled complexity hierarchy, in which representational capacity, depth, and regularization mechanisms are progressively increased while keeping the input resolution and training protocol fixed.

Model A is a deliberately lightweight architecture intended as a minimal baseline. It relies on a small number of depthwise-separable convolutional layers followed by max-pooling and a shallow fully connected head. The use of separable convolutions significantly reduces the number of trainable parameters and computational cost, enforcing a strong inductive bias toward learning coarse, low-level spatial patterns. A global average pooling layer is employed before the dense classifier, further limiting the model's capacity to memorize spatially localized features. As a result, model A primarily captures global shape cues rather than fine-grained gesture details.

Model B represents a modest increase in expressiveness and corresponds to a conventional shallow CNN. It employs standard convolutional layers followed by max-pooling and a fully connected classifier operating on flattened feature maps. Compared to model A, this design increases parameter count and spatial flexibility but lacks explicit architectural regularization mechanisms beyond implicit constraints imposed by depth and pooling. This makes model B more expressive but also more sensitive to distributional shifts, as later evidenced in the out-of-distribution evaluation.

Model C is a custom-designed architecture optimized for stable training and robust feature extraction under constrained data regimes. It combines depthwise separable convolutions, residual connections, and progressive spatial downsampling within a fully convolutional backbone. The network is organized into stages operating at decreasing spatial resolutions (48×48, 24×24, 12×12), allowing it to learn hierarchical representations ranging from local edge-like features to more abstract, gesture-level patterns. A global average pooling head followed by a compact dense layer ensures that classification decisions are based on aggregated semantic information rather than localized activations.

2.2 Architectural Design Choices and Technical Rationale

Several design decisions were made to ensure training stability, fair comparison, and robustness, particularly given the relatively small size and homogeneity of the training dataset.

First, depthwise separable convolutions are employed extensively in models A and C. This choice reduces parameter count while preserving the ability to model spatial structure, making it well-suited for scenarios where overfitting is a concern. In model C, separable convolutions are combined with residual connections, allowing gradients to propagate effectively through deeper stacks of layers and mitigating degradation effects commonly observed in deeper CNNs.

Second, Layer Normalization (LN) is used instead of Batch Normalization in model C. Since training is performed with relatively small batch sizes and with data augmentation enabled, LN provides more stable normalization statistics that are independent of batch composition. This choice improves convergence consistency and reduces sensitivity to batch-size variations during both tuning and final training.

Third, across all architectures (models A, B, and C), progressive spatial downsampling is implemented via strided convolutions rather than aggressive pooling. This preserves more information during early stages while still reducing computational complexity, enabling the networks to retain discriminative spatial cues relevant for fine-grained hand gesture recognition.

At the classification stage, all models adopt global average pooling in place of flattening, enforcing translation invariance and reducing the risk of overfitting by preventing the classifier from relying on specific spatial positions. In addition, a modest dropout rate is applied to the final dense layer across architectures to further discourage co-adaptation of high-level features.

Finally, label smoothing is applied uniformly across all models. This choice regularizes the softmax outputs by preventing overconfident predictions, which is particularly important when models are evaluated under domain shift. In model C, the option to initialize output biases using empirical class priors further stabilizes early training dynamics and reduces class collapse effects observed in simpler architectures.

Training optimization and regularization. Across all architectures, optimization is performed with the Adam optimizer. Training is regularized through early stopping monitored on validation accuracy and an adaptive learning-rate schedule (ReduceLROnPlateau), which lowers the learning rate when validation performance stagnates. Together, these mechanisms improve training stability and convergence behaviour, while limiting unnecessary computation once progress plateaus.

Model capacity and parameterization. To make the notion of "incremental complexity" explicit, the architectures are also compared in terms of trainable parameter count. Model A is a lightweight baseline with 2,814 trainable parameters, reflecting the strong inductive bias induced by separable convolutions and a compact classification head. Model B substantially increases capacity (249,331 trainable parameters), mainly due to the `Flatten+Dense` stage operating on high-dimensional feature maps. Model C achieves a favorable capacity–efficiency trade-off with 45,523 trainable parameters by relying on depthwise-separable convolutions, residual blocks, and global average pooling, which together reduce parameter growth while enabling deeper hierarchical feature extraction.

2.3 Hyperparameter Tuning and Final Model Selection

Hyperparameter tuning was carried out through a constrained grid search applied consistently across all three architectures (Models A, B, and C), exploring a small set of high-impact training hyperparameters: learning rate, batch size, and the use of data augmentation. For each architecture, all combinations in the grid were trained using the same stratified train–validation split, and model selection was based on validation accuracy computed on the held-out validation set. Early stopping was employed during each run to prevent overfitting and to limit unnecessary computation once validation performance plateaued, enabling an efficient yet systematic exploration of the search space. All trials were logged in a dedicated results table (`tuning_results.csv`), ensuring full reproducibility and allowing direct comparison across architectures under matched tuning conditions. The best configurations identified for each model are summarized in Table 1.

Table 1: *Best tuning configurations per model, reporting the best run with and without augmentation.*

Model	Learning rate	Batch size	Augment	Val. accuracy
A	5×10^{-4}	16	No	0.9087
A	1×10^{-3}	16	Yes	0.8584
B	1×10^{-3}	16	No	0.9772
B	1×10^{-3}	16	Yes	0.9840
C	3×10^{-4}	16	No	0.9977
C	5×10^{-4}	16	Yes	0.9954

The final model was selected as the overall best-performing configuration across the entire grid: Model C trained with learning rate 3×10^{-4}, batch size 16, and augmentation disabled achieved the highest validation accuracy observed during tuning. This best configuration was subsequently used for a clean final training run with checkpointing and evaluation artefacts generated accordingly. A detailed comparison of validation performance across architectures and the resulting model ranking is presented in Section 3.

3 Evaluation and Analysis

Building on the architectural comparison introduced in Section 2, this section evaluates how increasing model complexity translates into empirical performance differences.

This section presents the results obtained from the training and evaluation pipeline, focusing on the outputs that are most informative for assessing model performance, stability, and generalization. Although the experimental framework produces a broad set of diagnostic artifacts during development, only a selected subset is reported here in order to avoid redundancy and to emphasize interpretability.

In particular, the results include:

i. a quantitative comparison across architectures based on validation performance, including accuracy, precision, recall, and F1-score, used to support model selection (Section 3.1);

ii. an analysis of training dynamics through loss and accuracy curves on both training and validation sets, providing insight into convergence behaviour and learning stability (Section 3.2);

iii. a qualitative analysis of misclassified validation examples, aimed at identifying systematic error patterns and model-specific failure modes (Section 3.3);

iv. an integrated assessment of overfitting and underfitting, based on performance metrics, learning curves, and error structure (Section 3.4).

As noted above, additional outputs generated during training, such as intermediate diagnostics, repeated misclassification grids for non-selected models, and inspection-level evaluations, are not discussed in this section, as their primary role is to support debugging and model development rather than comparative performance analysis. All such artifacts are nonetheless retained in the project repository to ensure reproducibility and methodological transparency.

3.1 Performance Metrics

Model performance was assessed using standard multi-class classification metrics, including accuracy, precision, recall, and F1-score, computed on a held-out validation set. Each metric captures a different and complementary aspect of predictive behaviour, allowing a comprehensive view of both predictive quality and learning behaviour and a nuanced comparison across architectures beyond accuracy alone. While accuracy summarizes overall correctness, precision and recall capture complementary aspects of class-specific performance: precision reflects the model's ability to avoid false positive assignments, whereas recall measures its capacity to correctly retrieve all instances of a given class. The F1-score balances these two dimensions and is particularly informative when comparing models with different error profiles. Support values confirm that all metrics are computed on a nearly balanced validation set, ensuring that macro-averaged scores are meaningful and not driven by class imbalance.

Table 2: *Validation performance comparison across architectures. Metrics are macro-averaged over classes.*

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Model A	0.81	0.83	0.81	0.82
Model B	0.97	0.97	0.97	0.97
Model C	0.99	0.99	0.99	0.99

As shown in Table 2, a clear and monotonic improvement emerges across architectures. The baseline model (model A) exhibits limited accuracy and macro F1-score, indicating that its representational capacity is insufficient to reliably discriminate among the three gesture classes. This limitation manifests primarily through uneven recall across classes, suggesting that the model systematically fails to recognize certain gestures even when they are present in the input. In practical terms, this behaviour corresponds to a high rate of missed detections for specific classes, despite a moderate ability to correctly label some of the predictions it does make.

Model B marks a substantial qualitative shift in performance. Both precision and recall increase consistently across all classes, resulting in a strong improvement in the macro F1-score and overall accuracy. This indicates that the model not only reduces false positives, but also significantly mitigates false negatives. The balanced improvement across metrics suggests that the architectural enhancements introduced in model B enable the extraction of more stable and discriminative features, leading to a more symmetric and reliable error profile.

The most expressive architecture (model C) further consolidates these gains, achieving uniformly high precision and recall for all classes. The saturation of the F1-score reflects near-complete class separability within the validation distribution, with residual errors becoming rare and non-systematic. Importantly, the concurrent improvement in all reported metrics indicates that performance gains are not driven by a trade-off between precision and recall, but rather by a genuine reduction in both types of errors. This behaviour provides strong evidence that model C learns robust class-specific representations and generalizes effectively within the in-distribution validation setting.

Taken together, these results demonstrate that increasing architectural expressiveness translates into progressively better error control, moving from partial and class-dependent recognition in model A to stable and near-perfect classification in model C. This progression provides a principled justification for selecting model C as the final architecture for subsequent generalization analyses.

Prediction confidence on the validation set. In addition to accuracy-based metrics, the distribution of predicted probabilities provides a coarse *proxy* of confidence calibration. On the validation set, the mean maximum softmax probability increases with model expressiveness (A: 0.59, B: 0.88, C: 0.93), while class-averaged predicted probabilities remain approximately balanced across classes, suggesting that performance gains are not driven by degenerate class priors. This diagnostic is later used to interpret robustness under domain shift (Section 4).

Accordingly, for completeness, a summary of model capacity and confidence-related diagnostics is reported in Table 3.

Table 3: *Model capacity and confidence-related diagnostics on the validation set.*

Model	Trainable parameters	Model size (trainable)	Val max-prob ($\mu \pm \sigma$)
Model A	2,814	11 KB	0.59 ± 0.15
Model B	249,331	974 KB	0.88 ± 0.11
Model C	45,523	178 KB	0.93 ± 0.08

3.2 Training Dynamics

Training dynamics were analysed by jointly examining the evolution of loss and accuracy on both the training and validation sets. These curves are reported in Figures 1 and 2 and provide insight into how each architecture learns over time, whether performance improvements arise from genuine feature learning or from overfitting, and how effectively optimization translates into generalization within the validation distribution.

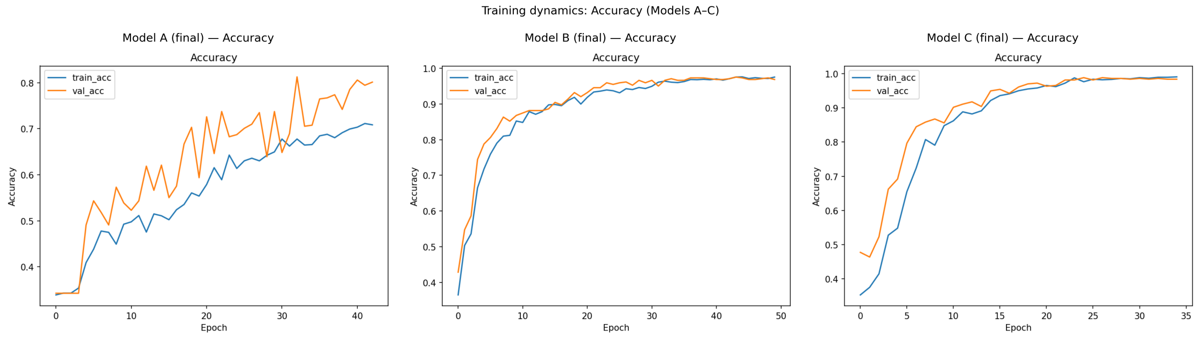

Figure 1: *Training and validation accuracy curves for the three final architectures.*

For the baseline architecture A, both training and validation loss decrease smoothly during the initial epochs but quickly reach a plateau, while accuracy saturates at relatively low values. The absence of a widening gap between training and validation curves indicates that the model does not overfit the data. Instead, the early stagnation of performance suggests limited representational capacity: the model is unable to further reduce classification error even on the training set. In practical terms, this behaviour is characteristic of mild underfitting, where the architecture captures only coarse visual patterns and fails to exploit additional training iterations to improve class discrimination.

Model B exhibits a markedly different learning trajectory. Loss decreases steadily across epochs for both training and validation sets, accompanied by a rapid and sustained increase in accuracy. The close alignment between training and validation curves throughout the optimization process indicates stable learning and efficient use of model capacity. From a practical perspective, this pattern suggests

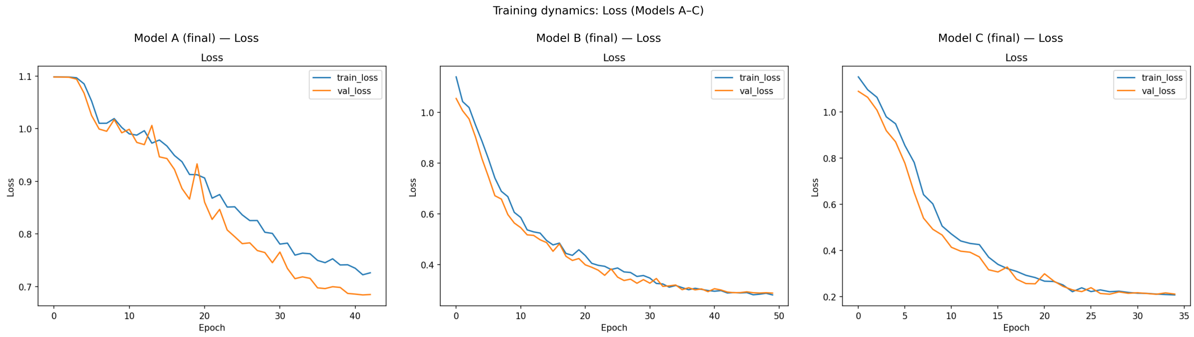

Figure 2: *Training and validation loss curves for the three final architectures.*

that the architectural enhancements introduced in model B enable the network to learn more expressive features without sacrificing generalization, leading to consistent improvements in both error reduction and predictive accuracy.

Model C shows similarly smooth and stable convergence, but with faster attainment of high accuracy and lower final loss values. Training and validation curves remain tightly coupled, with only a minimal and non-systematic generalization gap. This behaviour indicates that the increased model capacity is effectively controlled through regularization, allowing the network to fully exploit its expressiveness without overfitting. In practice, this translates into rapid learning of highly discriminative representations and stable convergence toward an optimal solution.

Overall, the progression of training dynamics across models mirrors the performance trends observed in the classification metrics. Model A is constrained by insufficient capacity, model B achieves an effective balance between learning and generalization, and model C combines high expressiveness with stable optimization. These dynamics confirm that the observed performance differences are rooted in structural properties of the architectures rather than in optimization artefacts or training instability.

3.3 Analysis of Misclassified Examples

When informative, a qualitative analysis of misclassified examples was conducted to identify systematic error patterns and to complement the quantitative evaluation provided by aggregate performance metrics. In this context, *misclassified samples* are defined as validation images for which the predicted class label differs from the ground-truth label. For each such instance, the predicted class confidence is computed as the maximum softmax probability associated with the model's output. This allows misclassifications to be inspected not only in terms of correctness, but also in terms of model confidence.

In practice, misclassified samples are ranked according to their prediction confidence and a fixed number of the most confident errors is selected for qualitative inspection. This design choice focuses the analysis on the most informative failure cases, namely those in which the model assigns high confidence to an incorrect prediction. Such errors are particularly relevant, as they reveal systematic confusions in the learned representation rather than borderline or noisy instances. Representative misclassified examples for each final architecture are shown in Figures 3, 4 and 5 (models A–C), ordered by decreasing prediction confidence.

The qualitative patterns observed differ substantially across architectures. For Model A, misclassifications are relatively frequent and show limited structural coherence, consistently with the uneven class-wise recall reported in Section 3.1. In practical terms, the "missed detections" highlighted by the metrics manifest here as a systematic tendency to collapse paper and scissors into the rock class: among the most confident errors, many true paper and scissors instances are predicted as rock, often with high confidence, suggesting reliance on incomplete or misleading cues (e.g., coarse hand silhouette and background contrast) rather than robust class-defining features. Conversely, true rock appears only rarely in this subset, and when it is misclassified it tends to be predicted as scissors, indicating an asymmetric error profile where paper and scissors are the more challenging classes for this low-capacity representation. Overall, these high-confidence mistakes align with the quantitative results and training dynamics for Model A, providing further evidence of underfitting.

For model B, misclassified examples become markedly fewer and more structured. Errors are primarily associated with visually ambiguous instances, such as partial hand occlusions, non-standard finger extensions, or poses captured at atypical orientations. Importantly, these misclassifications tend to occur at lower confidence levels, suggesting that the model exhibits increased uncertainty when confronted with borderline cases rather than systematically misinterpreting them. This pattern reflects a more refined internal representation of gesture classes and aligns with the balanced precision–recall improvements observed in the quantitative evaluation

Misclassifications for model C are rare and highly localized. When present, they almost exclusively involve intrinsically ambiguous images in which class boundaries are visually blurred, for example due to motion blur, extreme perspective, or transitional hand poses between gestures. Even in these cases, predicted confidence values are generally lower than those observed for incorrect predictions in simpler architectures, indicating that the model is aware of its uncertainty. This behaviour suggests that residual errors are driven by limitations in the visual signal itself rather than by deficiencies in the learned representation

In light of the above, the qualitative analysis of misclassified examples reinforces the conclusions drawn from performance metrics and training dynamics. As model capacity increases, errors transition from frequent and unstructured (model A), to rare and ambiguity-driven (models B and C). This progression indicates that improvements in classification performance are accompanied by a qualitative shift in the nature of errors, from representational failures to irreducible visual ambiguity.

3.4 Overfitting and Underfitting Assessment

The presence or absence of overfitting and underfitting was assessed by jointly analysing quantitative performance metrics, training and validation learning curves, and the qualitative structure of classification errors. This combined perspective allows potential optimization artefacts to be distinguished from genuine limitations in model capacity.

For the baseline architecture, evidence of underfitting is apparent. Training accuracy saturates at relatively low levels and does not continue to improve with additional epochs, while training and validation loss curves remain closely aligned. This pattern indicates that the model is unable to sufficiently reduce error even on the training data, ruling out overfitting and pointing instead to inadequate representational capacity. The qualitative analysis of misclassified examples further supports this diagnosis, as errors are frequent and lack a consistent structural pattern, suggesting incomplete feature learning rather than noise sensitivity.

In contrast, no evidence of overfitting is observed for the more expressive architectures. For both models B and C, training and validation accuracy curves evolve in parallel, and validation loss does not exhibit late-stage divergence or instability. The absence of a widening generalization gap indicates that performance improvements are not driven by memorization of the training data, but rather by the learning of representations that generalize reliably within the validation distribution. Model C, in particular, combines high predictive performance with stable optimization behaviour. The minimal and non-systematic differences between training and validation curves, together with the rarity and structure of residual errors, suggest that the model operates in a regime of effective capacity rather than over-parameterization. Consequently, the selected architecture demonstrates robustness under in-distribution conditions. More broadly, the comparison across models highlights the central role of architectural capacity in determining whether a model underfits, generalizes effectively, or risks overfitting.

4 Generalization Test

4.1 External In-Distribution Evaluation

In order to disentangle generalization within the same acquisition domain from robustness to genuine domain shift, the final model (model C) was additionally evaluated on the *rps-cv-images* dataset. This dataset is external to the training and validation splits, but shares similar characteristics in terms of background, lighting conditions, and gesture presentation.

The model achieves near-perfect performance on this benchmark, with an overall accuracy exceeding 99% across 2,188 test images and only a handful of isolated misclassifications. The confusion matrix is almost perfectly diagonal, and precision, recall, and F1-scores reach unity for all classes. Detailed quantitative results, including the classification report and confusion matrix, are provided in Appendix B.

These results confirm that the selected architecture generalizes extremely well to unseen data drawn from the same underlying distribution. Consequently, the performance degradation observed on the custom dataset can be attributed primarily to domain shift rather than to overfitting or instability of the learned representations.

4.2 External Out-of-Distribution Evaluation

Custom Dataset description

The custom dataset used for the out-of-distribution evaluation consists of a small collection of hand gesture images representing the three target classes (*rock*, *paper*, and *scissors*), acquired outside the original training distribution. The images were captured using consumer-grade devices under unconstrained conditions and include hands belonging to multiple individuals rather than a single subject. This choice introduces natural variability in hand shape, skin tone, finger proportions, and gesture execution style, thereby increasing the heterogeneity of the dataset.

In addition to subject variability, the images exhibit substantial diversity in background, lighting conditions, and viewpoint. Backgrounds range from uniform surfaces to textured environments such as wooden floors, tiled areas, and indoor furnishings, while lighting conditions vary from diffuse natural light to indoor artificial illumination. The hand poses are not strictly canonical: gestures may be partially rotated, cropped, or captured at non-standard distances and angles. Accessories such as watches or wristbands are also present in some images, further increasing visual complexity.

Although the dataset is approximately class-balanced, it is intentionally small and uncurated, reflecting realistic acquisition conditions rather than controlled laboratory settings. As such, it is not intended to provide statistically robust performance estimates, but rather to serve as a qualitative stress test for model robustness and generalization under domain shift. The dataset allows for a focused analysis of how well models trained on clean, curated data transfer to real-world scenarios characterized by higher intra-class variability and reduced visual consistency.

Out-of-Distribution Evaluation on Custom Data

The evaluation on the custom hand-gesture dataset highlights a clear and systematic difference in robustness across the three architectures under out-of-distribution conditions. Despite the dataset being nearly class-balanced, models A and B exhibit a pronounced bias towards the *scissors* class, as evidenced by highly skewed predicted class distributions and extreme mean predicted priors. In particular, model B collapses almost entirely onto *scissors* (predicted prior $\approx [0.003, 0.001, 0.996]$), while model A shows a slightly milder but still substantial preference for the same class.

This behaviour is reflected in their confusion matrices and per-class metrics, where *scissors* achieves comparatively high recall at the expense of *rock* and especially *paper*, whose recall drops to near-zero values. The corresponding misclassified-image grids further reveal that, under domain shift, visually ambiguous *rock* and *paper* gestures are consistently mapped to *scissors* with high confidence, indicating that simpler architectures tend to default to the most visually distinctive class when uncertainty increases.

Model C, while not immune to domain shift, substantially mitigates this failure mode. Its predictions are distributed across all three classes, yielding higher overall accuracy and macro F1-score, and a confusion matrix that preserves meaningful structure rather than degenerating into a single-class predictor. Although the mean predicted prior for *scissors* is close to zero before recalibration, the confusion matrix confirms that *scissors* is still actively predicted, albeit with lower confidence, pointing to a calibration issue rather than a hard classification collapse. The systematic class collapse observed for models A and B, and its mitigation in model C, are clearly illustrated by the confusion matrices reported in Appendix C (Figure 7).

Overall, these results suggest that increased representational capacity and architectural regularization improve robustness to real-world variability, while also highlighting the intrinsic difficulty of reliably distinguishing hand gestures under uncontrolled acquisition conditions. This behaviour is further characterized by the mean predicted class priors reported in Appendix C (Table 5).

5 Conclusion and Future Work

This project investigated the effectiveness of convolutional neural networks of increasing architectural complexity for the classification of hand gesture images representing rock, paper, and scissors. Through a systematic comparison of three CNN architectures trained under identical optimization settings, the analysis demonstrated a clear relationship between model capacity and predictive performance. While the baseline architecture exhibited signs of underfitting and limited discriminative power, progressively deeper and better-regularized models achieved very strong in-distribution performance, confirming the benefits of increased representational expressiveness for this task.

Beyond in-distribution evaluation, the study placed particular emphasis on model robustness under domain shift by introducing an external test set composed of unconstrained, real-world hand gesture images. This generalization test revealed that simpler architectures tend to collapse toward visually dominant classes when confronted with unfamiliar acquisition conditions, leading to highly skewed predictions and poor class balance. In contrast, the most expressive model retained a more structured decision behaviour and substantially mitigated this failure mode, although some degradation in confidence calibration remained. These findings highlight that strong validation performance alone is not sufficient to guarantee robustness in real-world scenarios and underscore the importance of explicitly evaluating models under out-of-distribution conditions.

Despite these encouraging results, several limitations must be acknowledged. The custom dataset used for the generalization test is intentionally small and qualitative in nature, preventing statistically robust performance estimates. Moreover, all models were trained exclusively on a curated dataset, without the use of advanced data augmentation strategies or domain adaptation techniques that could further enhance robustness. Future work could therefore explore the impact of more aggressive augmentation pipelines, the inclusion of real-world images during training, or transfer learning approaches to improve generalization. Additionally, investigating probabilistic calibration methods and confidence-aware decision mechanisms could help address residual uncertainty when models are deployed in unconstrained environments.

Appendix

A Additional Qualitative Diagnostics

A.1 Misclassified examples

Images are ranked by decreasing prediction confidence. Each panel reports the true label (T), the predicted label (P), and the associated softmax confidence score.

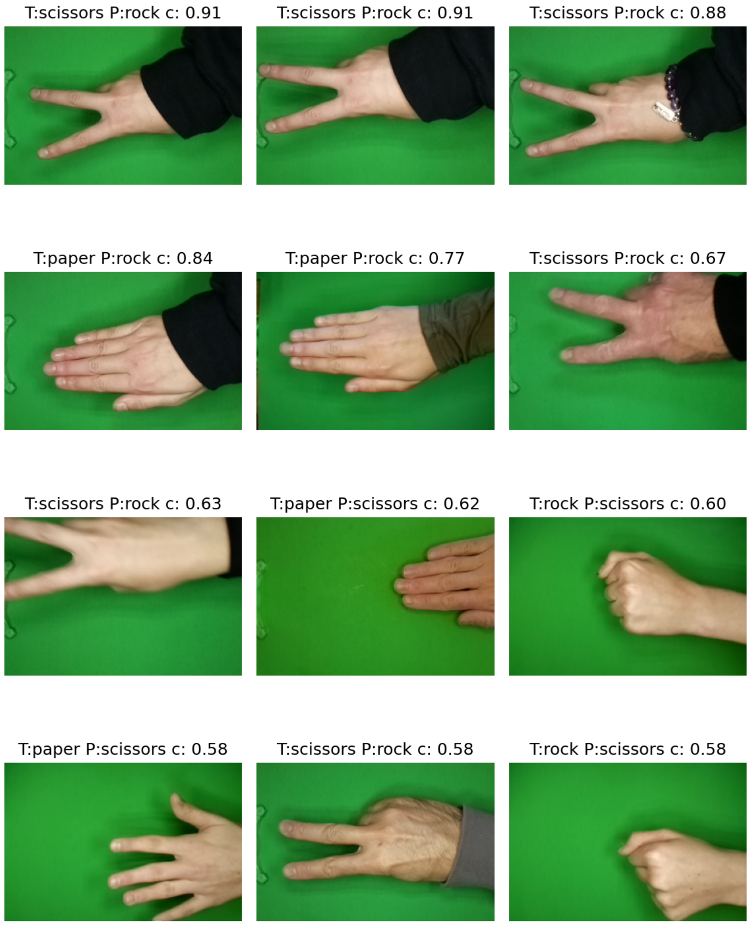

Figure 3: *Most confident misclassified validation samples for **Model A**.*

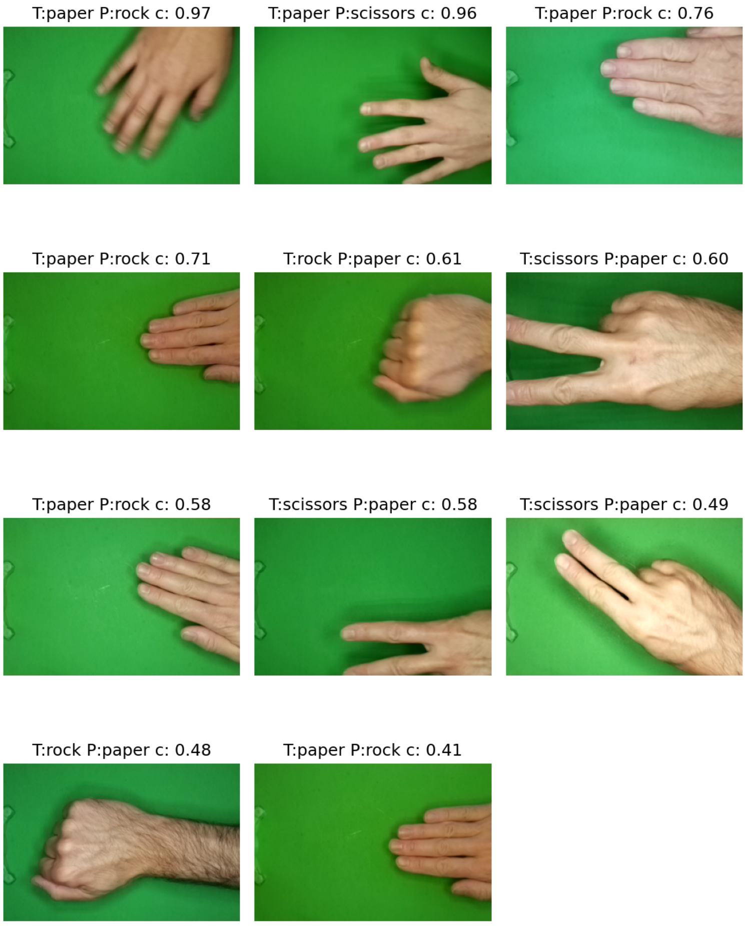

Figure 4: *Most confident misclassified validation samples for **Model B**.*

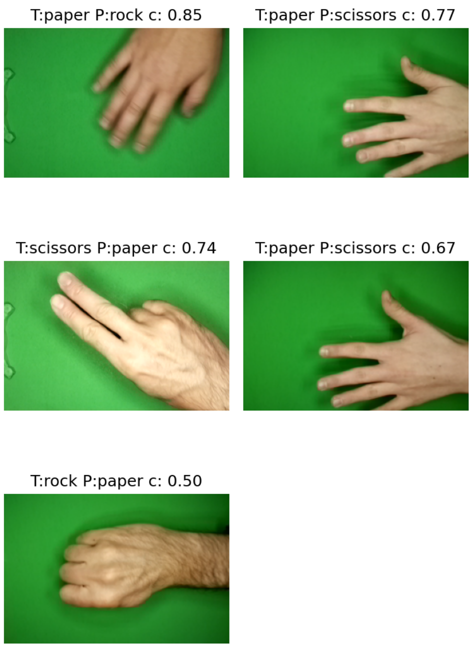

Figure 5: *Most confident misclassified validation samples for **Model C**.*

B External In-Distribution Evaluation Details

B.1 Classification report

Table 4: *Classification report for model C on the external in-distribution test set* rps-cv-images *(rounded values).*

Class	**Precision**	**Recall**	**F1-score**	**Support**
Rock	1.00	1.00	1.00	726
Paper	1.00	1.00	1.00	712
Scissors	1.00	1.00	1.00	750
Accuracy			1.00	2188
Macro avg	1.00	1.00	1.00	2188
Weighted avg	1.00	1.00	1.00	2188

B.2 Confusion matrix

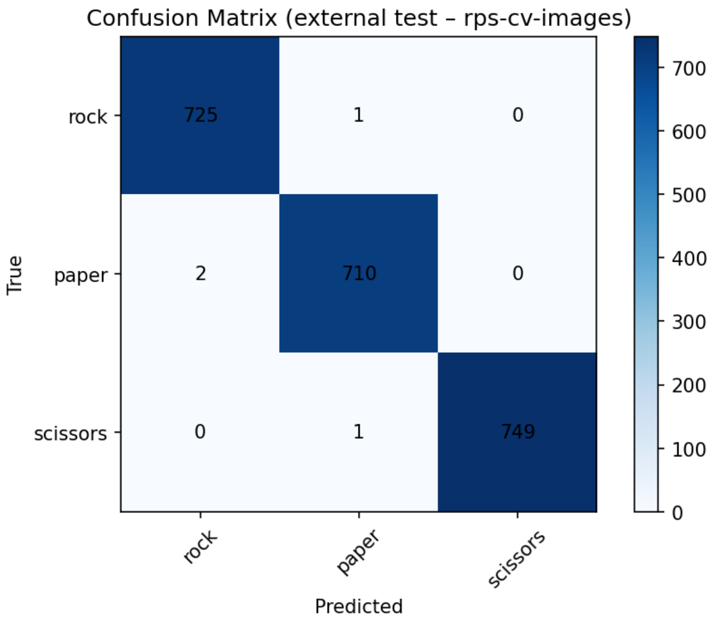

Figure 6: *Confusion matrix for model C on the external in-distribution test set (*rps-cv-images*).*

C External Out-Of-Distribution Evaluation Details

C.1 Confusion matrices

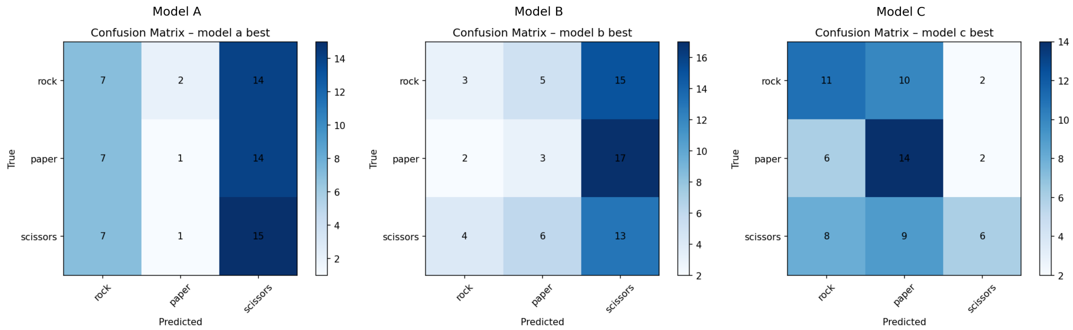

Figure 7: *Confusion matrices for the three architectures evaluated on the custom out-of-distribution dataset.*

C.2 Predicted class priors

Table 5: *Mean predicted class priors (pre-recalibration) for the three architectures on the custom out-of-distribution dataset.*

Model	**Rock**	**Paper**	**Scissors**
Model A	0.066	0.074	0.860
Model B	0.003	0.001	0.996
Model C	0.332	0.668	0.000
Target prior	0.333	0.333	0.333